Chapter 3. Coordinate and Time Systems

3.1. Historical Background

- Newton's Law of Gravitation

$$\vec{F}_g = -\frac{Gm_\oplus m_{sat}}{r^3}\vec{r}$$

  - $\mu = G(m_\oplus + m_{sat})$; $Gm_\oplus = 3.986\ 004\ \ 415 \times 10^5\, km^3 / sec^2$
  - $G = (6.673 \pm 0.001) \times 10^{-20}\, km^3 / kg \cdot sec^2$
  - $m_\oplus$; $5.973\ 332 \times 10^{24}\, kg$

- Gaussian Gravitational Constant $k\ @\sqrt{Gm_{sun}} = 0.017\ 202\ \ 098\ 95$
  - From the Earth's revolution around the Sun

$$r_\oplus \omega^2 = \frac{G(m_{sun} + m_\oplus)}{r_\oplus^2}$$

$$\Rightarrow \quad \omega^2 = \frac{G(m_{sun} + m_\oplus)}{r_\oplus^3} \qquad \leftarrow \omega = \frac{2\pi}{P_\oplus}$$

$$\Rightarrow \quad \frac{4\pi^2}{P_\oplus^2} = \frac{G(m_{sun} + m_\oplus)}{r_\oplus^3}$$

$$\Rightarrow \quad P_\oplus = 2\pi\sqrt{\frac{r_\oplus^3}{G(m_{sun} + m_\oplus)}} \qquad \leftarrow k\ @\sqrt{Gm_{sun}}$$

$$\Rightarrow \quad k = \frac{2\pi}{P_\oplus}\sqrt{\frac{r_\oplus^3}{1 + m_\oplus / m_{sun}}}$$

  - Units

$$P_\oplus = 365.256\ 383\, days$$
$$r_\oplus = 1.0\, AU$$
$$m_{sun} = 1.0$$
$$m_\oplus = 1 / 354{,}710\, solar\ masses$$

3.2. The Earth

- Basic Physical Parameters
  - Mean Equatorial Radius

    $R_{\oplus} = 6{,}378.136\ 3km$

  - Mean Polar Radius

    $b_{\oplus} = 6{,}356.751\ 600km$

  - Eccentricity or Flattening

    $$f = 0.003\ 352\ 813\ 178 = \frac{1}{298.257} = \frac{a-b}{a}$$
    $$e_{\oplus} = 0.081\ 819\ 221\ 456$$

  - Earth's Rotational Velocity

    $\omega_{\oplus} = 7.292\ 115 \times 10^{-5} \pm 1.5 \times 10^{-12} rad / sec$

  - Earth's Gravitational Parameter

    $\mu = 3.986\ 004\ 415 \times 10^{5} km^{3} / sec^{2}$

- Location Parameters=Terrestrial Parameters
  - Longitude (λ): Greenwich Meridian is the zero point
  - Latitude (ϕ): Reference the Earth's Equator

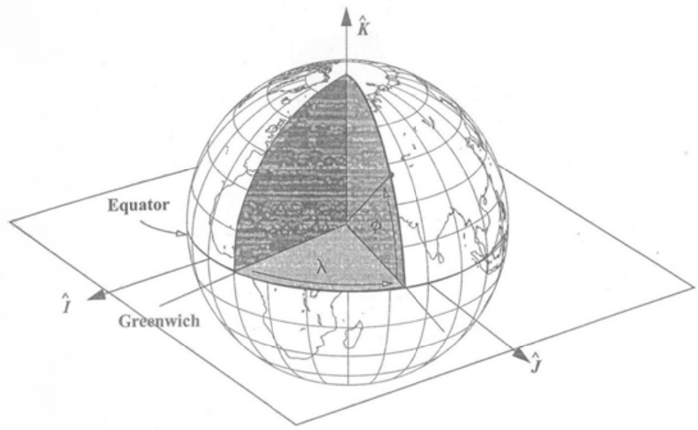

Figure 3-1. **Using Latitude and Longitude to Measure Location.** Terrestrial latitude, ϕ, and longitude, λ, reference the Earth's equator. The Greenwich meridian is the zero point for longitude, and the Earth's equator is the reference point for latitude.

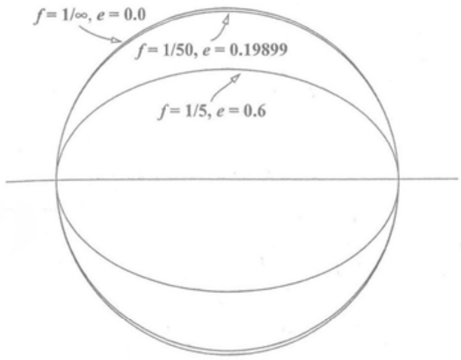

Figure 3-2. **Ellipsoidal Representation of the Earth.** We can approximate the Earth's shape with a flattening of about 1/298.3 to account for the 21 km difference in equatorial and polar radii. This flattening would be indistinguishable from the outer circle at this scale. Notice how closely the ellipse approaches a circle even with $f = 1/50$. For the JGM-3 model, the eccentricity of the Earth is 0.081 819 221 456.

- Shape of the Earth
  - Spherical Earth: the simplest representation (which suffices for many studies)
  - Reference Ellipsoid

    $$a_\oplus = R_\oplus = 6{,}378.136\ 3 km$$
    $$e_\oplus = 0.081\ 819\ 221\ 456$$

  - Geoid
    - Actual Mean Sea Level Surface
    - Deviates from the reference ellipsoid because of the uneven distribution of the Earth's mass

- No confusion with longitude (λ) for any Earth model
- A variety of latitude (ϕ) due to oblateness
  - ϕ_{gc} : Geocentric Latitude
  - ϕ_{gd} : Geodetic Latitude (Angle between Equatorial Plane and Normal to the surface of Ellipsoid)
  - ϕ_{as} ; ϕ_{gd} : Astronomical Latitude (Angle between Equatorial Plane and Normal to the surface of the Geoid)
  - ϕ_{rd} : Reduced Latitude (Auxiliary Parameter used for derivational purpose)

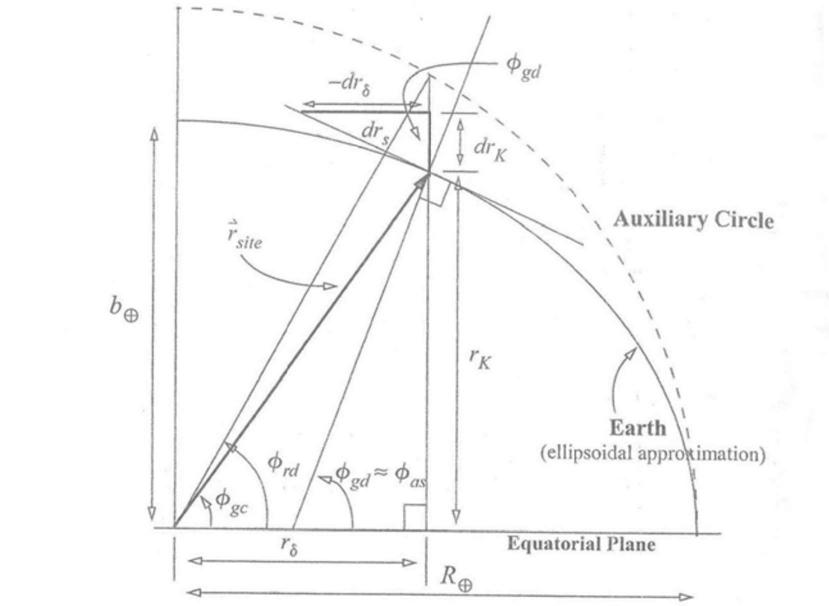

Figure 3-3. Latitude Geometry (exaggerated scale). The geodetic latitude, ϕ_{gd}, makes an angle perpendicular to the surface and the equatorial plane, whereas the geocentric latitude, ϕ_{gc}, is referenced to the center of the Earth. Astronomic latitude, ϕ_{as}, is very close to ϕ_{gd}. The center of the ellipsoid differs slightly for ϕ_{gd} and ϕ_{as}. Reduced latitude, ϕ_{rd}, is used only to derive relations.

- Conversion between each latitude

$$\vec{r}_{site} = \begin{bmatrix} r_i \\ r_j \\ r_k \end{bmatrix} = \begin{bmatrix} r_{site}\, cos\,\phi_{gc}\, cos\,\lambda \\ r_{site}\, cos\,\phi_{gc}\, sin\,\lambda \\ r_{site}\, sin\,\phi_{gc} \end{bmatrix} \tag{1}$$

We are only interested in relation between latitudes

- $r_\delta, r_k = f\left(\phi_{gc}, r_{site}\right)$

$$\Rightarrow \begin{cases} r_\delta = \sqrt{r_i^2 + r_j^{\,2}} = r_{site}\, cos\,\phi_{gc} \\ r_k = r_{site}\, sin\,\phi_{gc} \end{cases} \tag{2}$$

- $r_\delta, r_k = f\left(\phi_{rd}\,;\, R_\oplus, e_\oplus\right)$

$$\Rightarrow \begin{cases} r_\delta = R_\oplus\, cos\,\phi_{rd} \\ r_k = R_\oplus \sqrt{1 - e_\oplus^{\,2}}\, sin\,\phi_{rd} \\ r_{site} = \sqrt{r_\delta^{\,2} + r_K^{\,2}} = R_\oplus \sqrt{1 - e_\oplus^{\,2}\, sin^2\,\phi_{rd}} \end{cases} \quad \Leftarrow y_e = y_c \sqrt{1 - e^2} \tag{3~5}$$

- $\phi_{gd} = f(\phi_{rd})$

$$\Rightarrow \begin{cases} dr_\delta = -R_\oplus\, sin\,\phi_{rd}\, d\phi_{rd} \\ dr_k = R_\oplus \sqrt{1 - e_\oplus^2}\, cos\,\phi_{rd}\, d\phi_{rd} \\ dr_s = \sqrt{dr_\delta^{\,2} + dr_k^2} = R_\oplus \sqrt{1 - e_\oplus^2\, cos^2\,\phi_{rd}}\, d\phi_{rd} \end{cases}$$

$$\Rightarrow \begin{cases} sin\,\phi_{gd} = -\dfrac{dr_\delta}{dr_s} = \dfrac{sin\,\phi_{rd}}{\sqrt{1 - e_\oplus^2\, cos^2\,\phi_{rd}}} \\ cos\,\phi_{gd} = \dfrac{dr_k}{dr_s} = \dfrac{\sqrt{1 - e_\oplus^2}\, cos\,\phi_{rd}}{\sqrt{1 - e_\oplus^2\, cos^2\,\phi_{rd}}} \end{cases} \tag{A1~A2}$$

- $\phi_{rd} = f\left(\phi_{gd}\right)$

$$\oplus\begin{cases}\left(1-e_{\oplus}^{2}\right)sin^{2}\phi_{gd}=\dfrac{\left(1-e_{\oplus}^{2}\right)sin^{2}\phi_{rd}}{1-e_{\oplus}^{2}cos^{2}\phi_{rd}} & \Leftarrow\left(1-e_{\oplus}^{2}\right)(A1)^{2}\\ cos^{2}\phi_{gd}=\dfrac{\left(1-e_{\oplus}^{2}\right)cos^{2}\phi_{rd}}{1-e_{\oplus}^{2}cos^{2}\phi_{rd}} & \Leftarrow(A2)^{2}\end{cases}$$

$$\Rightarrow 1-e_{\oplus}^{2}sin^{2}\phi_{gd}=\frac{1-e_{\oplus}^{2}}{1-e_{\oplus}^{2}cos^{2}\phi_{rd}}$$

$$\Rightarrow 1-e_{\oplus}^{2}cos^{2}\phi_{rd}=\frac{1-e_{\oplus}^{2}}{1-e_{\oplus}^{2}sin^{2}\phi_{gd}} \qquad \text{(A3)}$$

From (A1)

$$sin\,\phi_{rd}=sin\,\phi_{gd}\sqrt{1-e_{\oplus}^{2}cos^{2}\phi_{rd}}$$

Introducing (A3)

$$sin\,\phi_{rd}=sin\,\phi_{gd}\sqrt{\frac{1-e_{\oplus}^{2}}{1-e_{\oplus}^{2}sin^{2}\phi_{gd}}} \qquad \text{(A4)}$$

From (A2)

$$cos\,\phi_{rd}=cos_{gd}\sqrt{\frac{1-e_{\oplus}^{2}cos^{2}\phi_{rd}}{1-e_{\oplus}^{2}}}$$

Introducing (A3)

$$cos\,\phi_{rd}=cos_{gd}\sqrt{\frac{1}{1-e_{\oplus}^{2}sin^{2}\phi_{gd}}} \qquad \text{(A5)}$$

- $r_{\delta},r_{k}=f\left(\phi_{gd};R_{\oplus},e_{\oplus}\right)$

Introducing (A5) into (3)

$$r_{\delta}=\frac{R_{\oplus}cos\,\phi_{gd}}{\sqrt{1-e_{\oplus}^{2}sin^{2}\phi_{gd}}}$$

Introducing (A4) into (4)

$$r_k = \frac{R_{\oplus}\left(1-e_{\oplus}^2\right) sin\,\phi_{gd}}{\sqrt{1-e_{\oplus}^2\, sin^2\,\phi_{gd}}}$$

Considering h_{ellp} for height above the ellipsoid

$$\begin{cases} r_{\delta} = \left(\dfrac{R_{\oplus}}{\sqrt{1-e_{\oplus}^2\, sin^2\,\phi_{gd}}} + h_{ellp} \right) cos\,\phi_{gd} \equiv \left(C_{\oplus} + h_{ellp}\right) cos\,\phi_{gd} \\ r_k = \left(\dfrac{R_{\oplus}\left(1-e_{\oplus}^2\right)}{\sqrt{1-e_{\oplus}^2\, sin^2\,\phi_{gd}}} + h_{ellp} \right) sin\,\phi_{gd} \equiv \left(S_{\oplus} + h_{ellp}\right) sin\,\phi_{gd} \end{cases}$$

$$\Rightarrow r_{site} = \sqrt{r_{\delta}^2 + r_k^2}$$

- $\phi_{gc} = f(\phi_{rd})$

From (3)-(4),

$$\begin{cases} sin\,\phi_{gc} = \dfrac{r_k}{r_{site}} = \dfrac{sin\phi_{rd}\sqrt{1-e_{\oplus}^2}}{\sqrt{1-e_{\oplus}^2\, sin^2\,\phi_{rd}}} \\ cos\,\phi_{gc} = \dfrac{r_{\delta}}{r_{site}} = \dfrac{cos\,\phi_{rd}}{\sqrt{1-e_{\oplus}^2\, sin^2\,\phi_{rd}}} \end{cases} \tag{6}$$

- $\phi_{rd} = f(\phi_{gc})$

From (6),

$$\begin{cases} sin\,\phi_{rd} = sin\,\phi_{gc}\sqrt{\dfrac{1-e_{\oplus}^2\, sin^2\,\phi_{rd}}{1-e_{\oplus}^2}} \\ cos\,\phi_{rd} = cos\,\phi_{gc}\sqrt{1-e_{\oplus}^2\, sin^2\,\phi_{rd}} \end{cases} \tag{7}$$

Introducing the following equation into (7)

$$\sqrt{\frac{1}{1-e_{\oplus}^2\, cos^2\,\phi_{gc}}} = \cdots = \sqrt{\frac{1-e_{\oplus}^2\, sin^2\,\phi_{rd}}{1-e_{\oplus}^2}}$$

$$\Rightarrow \sqrt{1-e_{\oplus}^2\, sin^2\,\phi_{rd}} = \sqrt{\frac{1-e_{\oplus}^2}{1-e_{\oplus}^2\, cos^2\,\phi_{gc}}}$$

We have

$$\begin{cases} sin\,\phi_{rd} = \dfrac{sin\,\phi_{gc}}{\sqrt{1-e_{\oplus}^{2}\,cos^{2}\,\phi_{gc}}} \\ cos\,\phi_{rd} = \dfrac{\sqrt{1-e_{\oplus}^{2}}\,cos\,\phi_{gc}}{\sqrt{1-e_{\oplus}^{2}\,cos^{2}\,\phi_{gc}}} \end{cases} \tag{8}$$

- $r_{\delta}, r_{k} = f\left(\phi_{gc};\, R_{\oplus}, e_{\oplus}\right)$

Introducing (8) into (3)

$$\begin{cases} r_{\delta}\ ;\ \left(\dfrac{R_{\oplus}\sqrt{1-e_{\oplus}^{2}}}{\sqrt{1-e_{\oplus}^{2}\,cos^{2}\,\phi_{gc}}} + h_{ellp}\right) cos\,\phi_{gc} \\ r_{k} = \left(\dfrac{R_{\oplus}\sqrt{1-e_{\oplus}^{2}}}{\sqrt{1-e_{\oplus}^{2}\,cos^{2}\,\phi_{gc}}} + h_{ellp}\right) sin\,\phi_{gc} \\ \\ r_{site} = \sqrt{r_{\delta}^{2} + r_{k}^{2}} \end{cases} \tag{9}$$

Note that (9) are approximate because h_{ellp} is not measured along the geocentric radius

- $\phi_{gc} \Leftrightarrow \phi_{gd}$

$$tan\,\phi_{rd} = \frac{sin\,\phi_{rd}}{cos\,\phi_{rd}}$$

$$\Rightarrow \begin{cases} tan\,\phi_{rd} = tan\,\phi_{gd}\sqrt{1-e_{\oplus}^{2}} & \Leftarrow (A4) \\ tan\,\phi_{rd} = \dfrac{tan\,\phi_{gc}}{\sqrt{1-e_{\oplus}^{2}}} & \Leftarrow (8) \end{cases}$$

$$\Rightarrow \begin{cases} tan\,\phi_{gd} = \dfrac{tan\,\phi_{gc}}{1-e_{\oplus}^{2}} \\ tan\,\phi_{gc} = \left(1-e_{\oplus}^{2}\right) tan\,\phi_{gd} \end{cases}$$

- Gravitational Model
  - Theoretical Gravity $g_{th} = f\left(\phi_{gd}, e_{\oplus}\right)$

$$g_{th} = g_{eq}\left(\frac{1+k_g \sin^2\phi_{gd}}{\sqrt{1-e_{\oplus}^2 \sin^2\phi_{gd}}}\right)$$

$$\begin{cases} g_{eq} = 9.780325335\, m/s^2 \\ g_{pole} = 9.832184937 m/s^2 \\ k_g = \dfrac{b_{\oplus} g_{pole}}{R_{\oplus} g_{eq}} - 1 = 0.001931852 \end{cases}$$

  - Mean value of theoretical gravity

$$\overline{g_{th}} = 9.797643222\, m/s^2$$

- Geopotential Surfaces
  - Geoid = Equipotential Surfaces

    = Potential of Mean Sea Level Surfaces at which the gravity is equal at all point

  - $N_{\oplus} = h_{ellp} - H_{MSL}$

$$\begin{cases} N_{\oplus} \text{ : geoid's undulation} \\ h_{ellp} \text{ : ellipsoid height} \\ H_{MSL} \text{ : actual height above the geoid} \end{cases}$$

$$N_{\oplus} = \frac{\mu}{r g_{th}} \sum_{l=2}^{\infty}\sum_{m=0}^{l}\left(\frac{R_{\oplus}}{r}\right)^{l} \overline{P}_{l,m}\left[\sin\phi_{gc}\right]\left\{\overline{C}_{l,m}\cos m\lambda + \overline{S}_{l,m}\sin m\lambda\right\}$$

$$\begin{bmatrix} r_i \\ r_j \\ r_k \end{bmatrix} = \left|r_{ijk}\right| \begin{bmatrix} \cos\phi_{gc}\cos\lambda \\ \cos\phi_{gc}\sin\lambda \\ \sin\phi_{gc} \end{bmatrix} = \begin{bmatrix} \left(C_{\oplus} + h_{ellp}\right)\cos\phi_{gd}\cos\lambda \\ \left(C_{\oplus} + h_{ellp}\right)\cos\phi_{gd}\sin\lambda \\ \left(S_{\oplus} + h_{ellp}\right)\sin\phi_{gd} \end{bmatrix}$$

3.3. Coordinate Systems

- Definition of Rectangular Coordinate System
  - Origin
  - Fundamental Plane
  - Preferred Direction

- Common Choices for the Origin of Coordinate Systems
  - Object's Center of Mass (CM)
  - System's Center of Mass (Barycenter)
  - Barycenter with Rotating Frame (Synodic System)

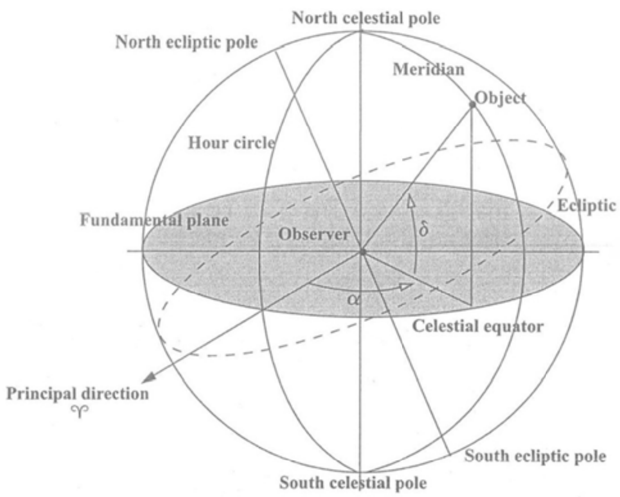

Figure 3-6. Geometry of the Celestial Sphere. The celestial sphere is based on an observer's perceived view of objects in space. A meridian, or hour circle, is any circle that passes through the observer at the center. I've shown the principal direction to help us define right ascension, α, and declination, δ.

- Geometry of Celestial Sphere (천구)
    - **Great Circles**: Intersection of the celestial sphere with any plane passing through the origin
    - **Celestial Poles (N&S)**: along the Earth's rotational axis

      Celestial Equator: Earth's equator onto celestial sphere
    - **Hour Circles = Meridian**: Great circles perpendicular to the celestial equator
    - **Ecliptic**: plane of the Earth's mean orbit about the Sun
    - **Obliquity (of the Ecliptic)**: angle between Earth's mean equator and ecliptic
    - **Line of Nodes**: line of intersection of equator and ecliptic
    - **Equinoxes**: Points of intersection of equator and ecliptic
        - **Vernal Equinox:** δ goes from (-) to (+)
        - **Autumnal Equinox**: δ goes from (+) to (-)
    - **Ecliptic Latitude & Longitude = Celestial Latitude & Longitude**
        - ϕ: ecliptic is the fundamental plane
        - λ: from the vernal equinox positive to the East
    - **Right Ascension (α) and Declination (δ)**
        - α: from the vernal equinox positive to the East on the equatorial plane
        - δ: measured northward from the equator to the object's location
    - **Hour Angle**: measured positive to the West
        - Local Hour Angle (LHA): Primary (Local, Observer) Hour Circle to the Hour Circle of an Object

◆ Greenwich Hour Angle (GHA)

◆ LHA = GHA + λ

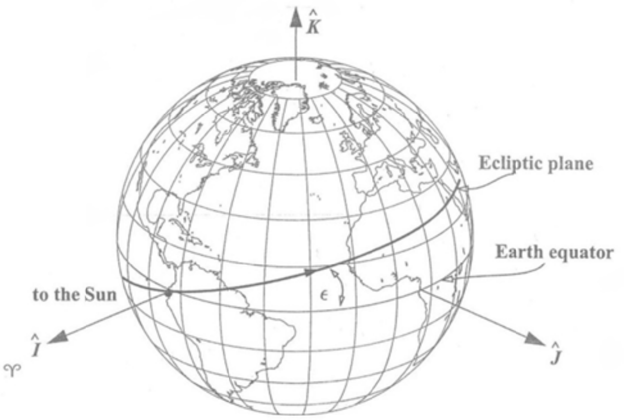

Figure 3-7. **Geometry of the Vernal Equinox.** The Earth's mean orbit around the Sun forms the ecliptic plane. The Earth's equatorial plane is inclined about 23.5° to the ecliptic. When the Sun is at the intersection of the two planes (has zero declination) and is at the ascending node, as viewed from Earth, it's the first day of spring.

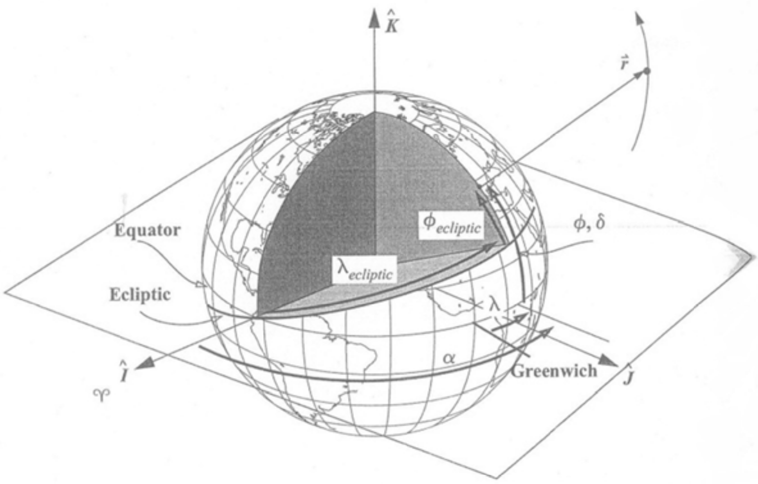

Figure 3-8. **Using Right Ascension and Declination or Longitude and Latitude to Measure Location.** Notice that both terrestrial latitude and longitude, ϕ and λ, and right ascension and declination, α and δ, reference the Earth's equator, whereas ecliptic latitude and longitude are celestial coordinates that reference the ecliptic. Declination and terrestrial latitude are the same, but right ascension and terrestrial longitude reference different starting locations.

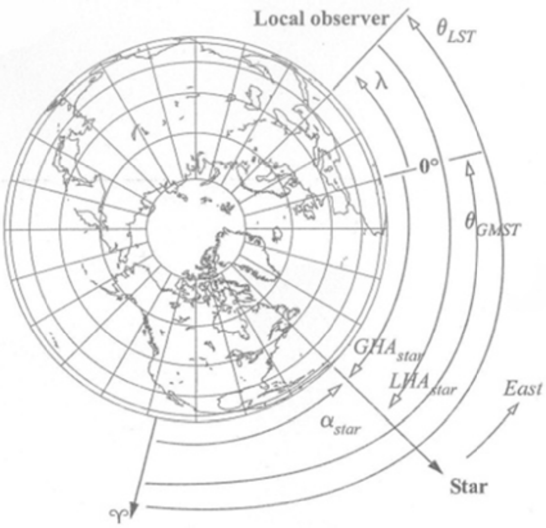

Figure 3-9. **Geometry for LHA and GHA.** GHA, and LHA are always measured positive to the west *from* the observer to the object of interest. Longitude, λ, is positive to the east. Both GHA_{star} and LHA_{star} are positive here. We use subscripts to avoid confusion when referring to specific objects. θ_{LST} and θ_{GMST} will be discussed shortly.

3.3.1. Interplanetary System

- Heliocentric Coordinate System
  - Mainly used for interplanetary missions
  - Origin is located at the center of the Sun
  - Ecliptic is the fundamental plane
  - Primary direction is along the line of nodes

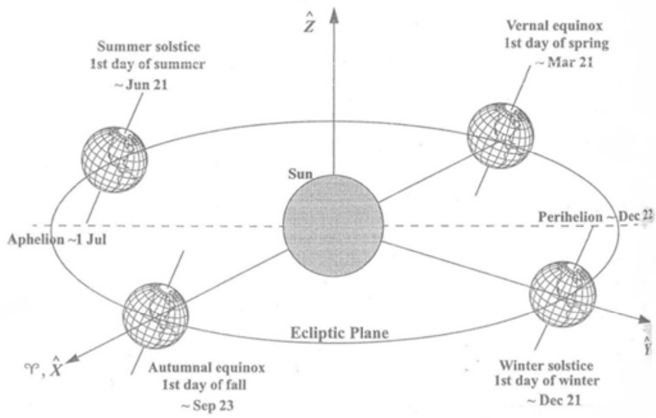

Figure 3-10. Heliocentric Coordinate System, XYZ. The primary direction for this system is the intersection of the ecliptic and equatorial planes. The positive sense (denoted ♈) is given by the location of the Sun with respect to the Earth as it crosses the equatorial plane on the first day of spring. All seasonal dates are for the Northern Hemisphere. The revolution of the Earth about the Sun doesn't match the Gregorian calendar exactly. Only after about 28 years are the dates approximately the same. This affects the dates for the seasons. Seidelmann (1992:477) lists the earliest seasonal dates that occur in 2096 (March 19, June 20, September 21, and December 20). The latest seasonal dates occurred in 1903 (March 21, June 22, September 24, and December 23).

- International Celestial reference System (Frame) = *ICRS* = *ICRF*
  - Origin is located at the Barycenter of the Solar System
  - Primary direction uses the IAU-76/FK5 J2000

3.3.2. Earth-based System

- Geocentric Equatorial Coordinate System (*IJK*)

  =Earth-Centered Inertial (*ECI*)

  =Conventional Inertial System (*CIS*)
  - Origin is located at the center of the Earth
  - Earth Equator is the fundamental plane
  - I-axis points toward vernal equinox, which is fixed at a specific epoch for most applications (often uses J2000)
  - Most common system in Astrodynamics

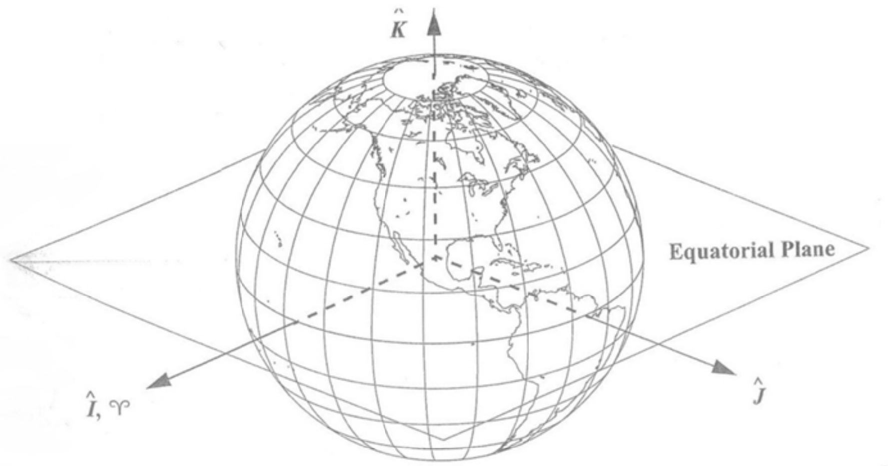

Figure 3-11. Geocentric Equatorial System (IJK). This system uses the Earth's equator and the axis of rotation to define an orthogonal set of vectors. The vernal equinox direction is fixed at a specific epoch for most applications.

- Geocentric Celestial Coordinate System

  = Geocentric Celestial Reference Frame (*GCRF*)

  - Standard Inertial Coordinate System
  - Geocentric counter part of *ICRF*

- Body-Fixed Coordinate System

  =International Terrestrial Reference Frame (*ITRF*)

  =Earth-Centered Earth Fixed (*ECEF*)

  - Use this system to process actual observations and acceleration calculations

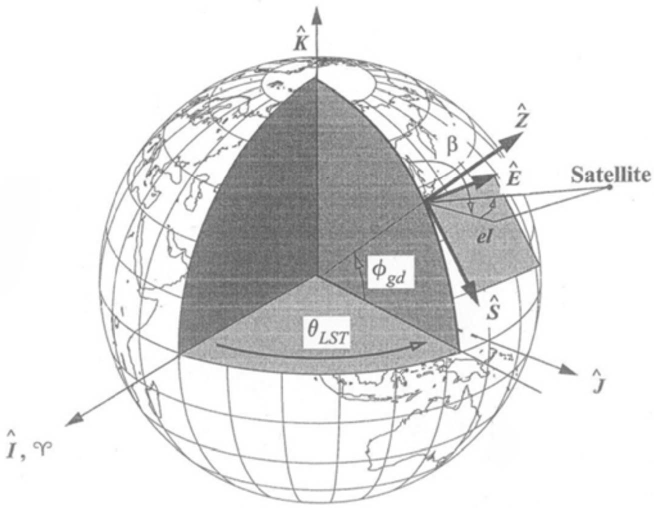

Figure 3-12. Topocentric Horizon Coordinate System, SEZ. This system moves with the Earth, so it's not fixed like the ITRF. θ_{LST} is required to orient the SEZ system to a fixed location. Note that for precise applications, θ_{AST} and longitude replace θ_{LST}. The SEZ system is used extensively in radar observations and often involves azimuth, β, and elevation, *el*. Geodetic latitude is common, ϕ_{gd}, but astronomical latitude is also used in many systems.

- Topocentric Horizontal Coordinate System (*SEZ*)
  - Useful in observing satellites
  - Extensively used with sensor systems
  - Rotates with the site
  - Local Horizontal is the fundamental plane
    - S-south
    - E-east
    - Z-local vertical
  - Longitude (or Hour Angle) and Latitude are required to locate and orient the SEZ-system
    - θ_{LST} / (θ_{AST} , λ)
    - ϕ_{gd} , ϕ_{as}
  - Defines 'Look Angles' to view the satellite

- Azimuth (β)
  - Horizontal Angle
  - Measured from the North, clockwise to the location beneath the object of interest
- Elevation (el)
  - Vertical Angle
  - Measure from the local horizon positive up to the object

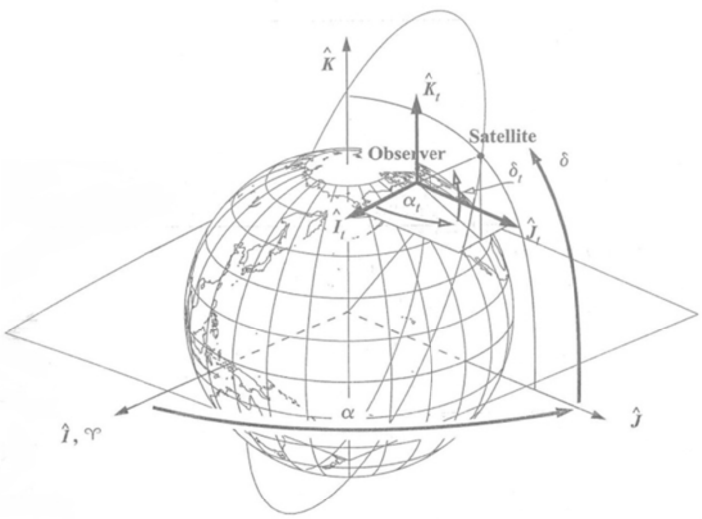

Figure 3-13. Topocentric Equatorial Coordinate System, $I_tJ_tK_t$. This system uses the orientation axes of the IJK system, but it's translated to a particular site. The origin rotates with the Earth but maintains the fixed orientation. The difference in numerical values (δ_t and δ) is apparent in the figure. The topocentric declination, δ_t, in this figure is about 20°, whereas the declination, δ, is about 50°.

- Topocentric Equatorial Coordinate System ($I_tJ_tK_t$)
  - Geocentric Equatorial Coordinate System with the origin translated from the Earth's center to the origin of the topocentric horizon system-the observer
  - Axis Orientation is the same as IJK-system

    It is parallel to the vernal equinox direction

3.3.3. Satellite-based System

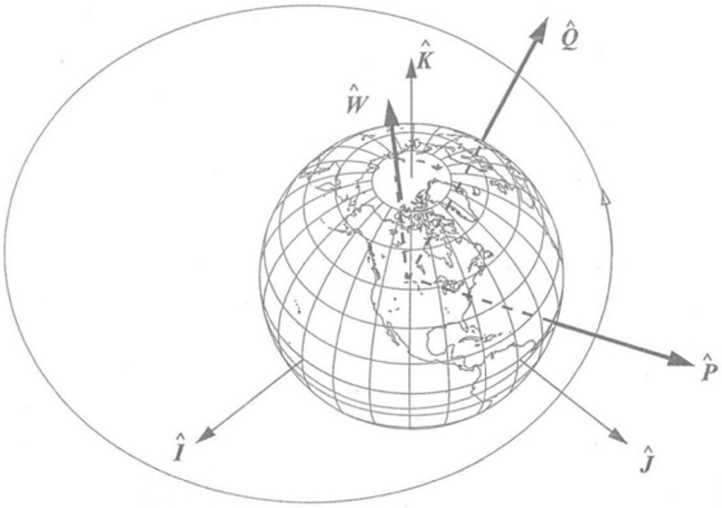

Figure 3-14. Perifocal Coordinate System, PQW. This system points towards perigee. Satellite motion is in the *P*-*Q* plane. The *W* axis is normal to the orbit plane. Although this particular orbit is near equatorial (the *J* axis is 10° up from the *P*-*Q* plane), it doesn't have to be. We often view this system looking down from the *W* axis.

- Perifocal Coordinate System (*PQW*)
  - Origin at the Earth (focus)
  - Satellite orbital plane is the fundamental plane
    - *P*: Perigee
    - *Q*: 90 degrees from the *P*-axis in the direction of motion
    - *W*: normal to the orbit in Right Hand System
  - Nodal Coordinate System
    - *P* is the direction of ascending node

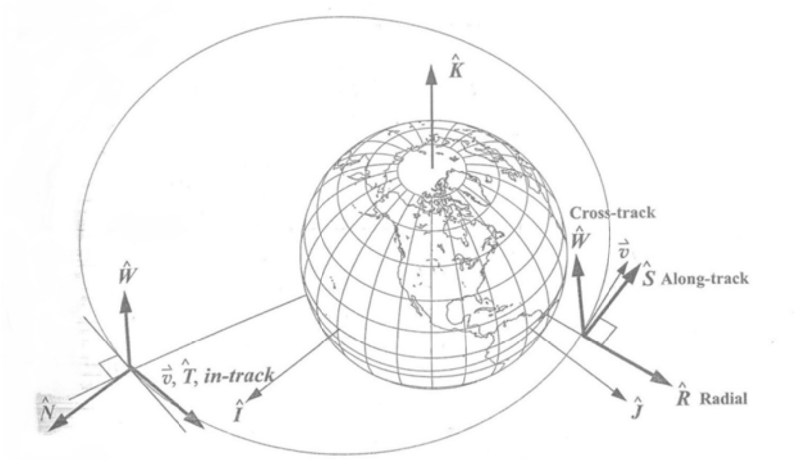

Figure 3-15. Satellite Coordinate Systems, RSW and NTW. These coordinate systems move with the satellite. The *R* axis points to the satellite, the *W* axis is normal to the orbital plane (and *not* usually aligned with the *K* axis), and the *S* axis is normal to the position vector and positive in the direction of the velocity vector. The *S* axis is aligned with the velocity vector *only* for circular orbits. In the NTW system, the *T* axis is always parallel to the velocity vector. The *N* axis is normal to the velocity vector and is *not* aligned with the radius vector, except for circular orbits, and at apogee and perigee in elliptical orbits.

- Satellite Coordinate System (*RSW*)

  =Gaussian Coordinate System(*RTN* = Radial, Transverse, Normal)

  - Moves with the Satellite
    - *R*: pointing from the Earth's center along the radius vector
    - *S*: Normal to R on the orbital plane
    - *W*: Perpendicular to the orbital plane in Right Hand System

    - Radial Position or Displacement
    - Along-Track/Transverse Displacement
    - Cross-track Position
  - *W* is not usually aligned with *K*-axis
  - *S* is aligned with velocity vector only for circular orbits

- Satellite Coordinate System (*NTW*)
  - Moves with the satellite
    - *N*: lies in the orbital plane, normal to the velocity vector
    - *T*: tangential to the orbit
    - *W*: perpendicular to the orbital plane in Right Hand System
  - In-Track/Tangential Displacement
    - Deviations along the *T*-axis
    - Not the same as along-track variations in the *RSW* system

- Equinoctial Coordinate system (*EQW*)
  - Orbital Plane is the fundamental plane
  - Start from *IJK*
    - *E*-axis is defined by $R_3(\Omega) \rightarrow R_1(i) \rightarrow R_3(-\Omega)$
    - *Q* lies in the orbital plane
    - *W* is perpendicular to the orbital plane in Right Hand System

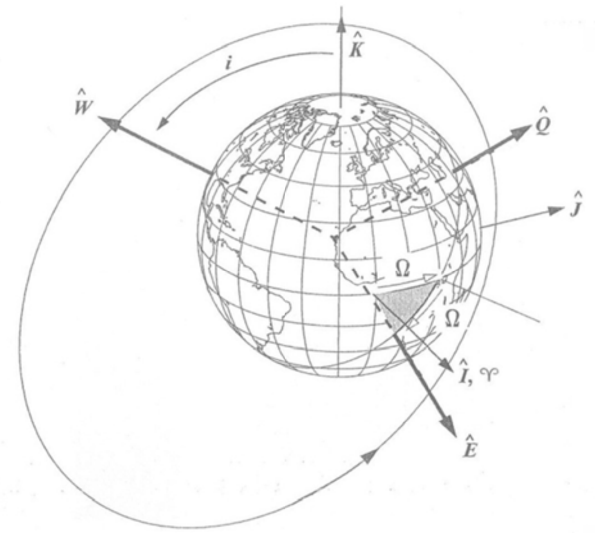

Figure 3-16. Equinoctial System, EQW. This system is formed starting at the IJK system (I axis). Rotating through the right ascension of the ascending node (Ω), tilting through the inclination (i), and then rotating back through Ω places us in the correct orientation and defines the E axis. The Q axis is perpendicular to the E axis in the orbital plane. The W axis is normal to the orbit.

TABLE 3-1. **Summary of Coordinate Systems.** With a few exceptions, most coordinate systems are simply variations of the ones in this table. The IJK letters indicate a generic Earth centered coordinate system. I'll use subscripts to indicate variations of the IJK coordinate system. The ITRF is defined for certain years, indicated by the ##—ITRF-94.

System	Symbol	Origin	Fundamental Plane	Principal Direction	Example Use
			Interplanetary systems		
Heliocentric	XYZ	Sun	Ecliptic	Vernal equinox	Patched conic
Solar system*	$(XYZ)_{ICRF}$	Barycenter	varies	varies	Planetary motion
			Earth-based systems		
Geocentric†	IJK	Earth	Earth equator	Vernal equinox	General
Earth	$(IJK)_{GCRF}$	Earth	varies	varies	Perturbations
Body-fixed	$(IJK)_{ITRF\text{-}\#\#}$	Earth	Earth equator	Greenwich meridian	Observations
Earth-Moon (synodic)	$(IJK)_S$	Barycenter	Invariable plane	Earth	Restricted three-body
Topocentric horizon	SEZ	Site	Local horizon	South	Radar observations
Topocentric equatorial	$(IJK)_t$	Site	Parallel to Earth equator	Vernal equinox	Optical observations
			Satellite-based systems		
Perifocal**	PQW	Earth	Satellite orbit	Periapsis	Processing
Satellite radial	RSW††	Satellite	Satellite orbit	Radial vector	Relative motion, Perturbations
Satellite normal	NTW	Satellite	Satellite orbit	Normal to velocity vector	Perturbations
Equinoctial	EQW	Satellite	Satellite orbit	Calculated vector	Perturbations

3.4. Coordinate Transformations

3.4.1 Rotational Transformations

- Rotational Matrix

  - $R_1(\alpha)=\begin{bmatrix}1 & 0 & 0\\ 0 & C_\alpha & S_\alpha\\ 0 & -S_\alpha & C_\alpha\end{bmatrix}$ about x-axis in RHS

    $R_2(\alpha)=\begin{bmatrix}C_\alpha & 0 & -S_\alpha\\ 0 & 1 & 0\\ S_\alpha & 0 & C_\alpha\end{bmatrix}$ about y-axis in RHS

    $R_3(\alpha)=\begin{bmatrix}C_\alpha & S_\alpha & 0\\ -S_\alpha & C_\alpha & 0\\ 0 & 0 & 1\end{bmatrix}$ about z-axis in RHS

  - (R_1, R_2, R_3) are orthogonal

    $\Leftrightarrow$ $R^{-1}=R^T$

    $\Leftrightarrow$ Rows/Columns are scalar components of mutually independent unit vectors

- Direction Cosine Matrix

  -

$$\begin{cases} \vec{A}^i = A_{i1}\hat{i}_1 + A_{i2}\hat{i}_2 + A_{i3}\hat{i}_3 \\ \vec{A}^s = A_{s1}\hat{s}_1 + A_{s2}\hat{s}_2 + A_{s3}\hat{s}_3 \end{cases}$$ same physical quantity with different representations

$$\begin{cases} A_{i_1} = \vec{A}^i \cdot \hat{i}_1 = \vec{A}^s \cdot \hat{i}_1 = \left(A_{s1}\hat{s}_1 + A_{s2}\hat{s}_2 + A_{s3}\hat{s}_3\right)\cdot \hat{i}_1 = A_{s1}\hat{s}_1 \cdot \hat{i}_1 + A_{s2}\hat{s}_2 \cdot \hat{i}_1 + A_{s3}\hat{s}_3 \cdot \hat{i}_1 \\ A_{i_2} = \vec{A}^i \cdot \hat{i}_2 = \vec{A}^s \cdot \hat{i}_2 = \left(A_{s1}\hat{s}_1 + A_{s2}\hat{s}_2 + A_{s3}\hat{s}_3\right)\cdot \hat{i}_2 = A_{s1}\hat{s}_1 \cdot \hat{i}_2 + A_{s2}\hat{s}_2 \cdot \hat{i}_2 + A_{s3}\hat{s}_3 \cdot \hat{i}_2 \\ A_{i_3} = \vec{A}^i \cdot \hat{i}_3 = \vec{A}^s \cdot \hat{i}_3 = \left(A_{s1}\hat{s}_1 + A_{s2}\hat{s}_2 + A_{s3}\hat{s}_3\right)\cdot \hat{i}_3 = A_{s1}\hat{s}_1 \cdot \hat{i}_3 + A_{s2}\hat{s}_2 \cdot \hat{i}_3 + A_{s3}\hat{s}_3 \cdot \hat{i}_3 \end{cases}$$

$$\Leftrightarrow \begin{pmatrix} A_{i1} \\ A_{i2} \\ A_{i3} \end{pmatrix} = \begin{bmatrix} \hat{i}_1 \cdot \hat{s}_1 & \hat{i}_1 \cdot \hat{s}_2 & \hat{i}_1 \cdot \hat{s}_3 \\ \hat{i}_2 \cdot \hat{s}_1 & \hat{i}_2 \cdot \hat{s}_2 & \hat{i}_2 \cdot \hat{s}_3 \\ \hat{i}_3 \cdot \hat{s}_1 & \hat{i}_3 \cdot \hat{s}_2 & \hat{i}_3 \cdot \hat{s}_3 \end{bmatrix} \begin{pmatrix} A_{s1} \\ A_{s2} \\ A_{s3} \end{pmatrix}$$

$\Uparrow$

Direction Cosine Matrix

$$\Leftrightarrow \vec{A}^i = R^{is}\vec{A}^s$$

- Direction Cosine Matrix is the component of one set of basis vectors with respect to the other set of basis vectors
- $\vec{A}$ is transformed from s-frame to i-frame

- Ultimately rotation matrix & direction cosine matrix are equivalent to each other, though the process is different

3.4.2 Common Transformations

- $IJK \Leftrightarrow SEZ$
  - Rotational Matrix

$$\begin{aligned} \vec{r}_{IJK} &= R_3\left(-\theta_{LST}\right)R_2\left(-\left(90^\circ - \phi_{gd}\right)\right)\vec{r}_{SEZ} \\ &= R_3^{\,T}\left(\theta_{LST}\right)R_2^{\,T}\left(90^\circ - \phi_{gd}\right)\vec{r}_{SEZ} \end{aligned}$$

  - Direction Cosine Matrix

$$\hat{Z} = \frac{\vec{r}_{site}}{\left|\vec{r}_{site}\right|}, \quad \hat{E} = \frac{\hat{k} \times \hat{Z}}{\left|\hat{k} \times \hat{Z}\right|}, \quad \hat{S} = \hat{E} \times \hat{Z}$$

$$\vec{r}_{IJK} = \begin{bmatrix} \hat{S} & \hat{E} & \hat{Z} \end{bmatrix} \vec{r}_{SEZ} \quad \Leftrightarrow \quad \vec{r}_{SEZ} = \begin{bmatrix} \hat{S}^T \\ \hat{E}^T \\ \hat{Z}^T \end{bmatrix} \vec{r}_{IJK}$$

- $IJK \Leftrightarrow PQW$
  - $$\begin{aligned} \vec{r}_{IJK} &= R_3(-\Omega) R_1(-i) R_3(-\omega) \vec{r}_{PQW} \\ &= R_3^T(\Omega) R_1^T(i) R_3^T(\omega) \vec{r}_{PQW} \end{aligned}$$
  - $$\hat{P} = \frac{\vec{e}}{|\vec{e}|}, \ \hat{W} = \frac{\vec{r} \times \vec{v}}{|\vec{r} \times \vec{v}|}, \ \hat{Q} = \hat{W} \times \hat{P}$$

    $$\vec{r}_{IJK} = \begin{bmatrix} \hat{P} & \hat{Q} & \hat{W} \end{bmatrix} \vec{r}_{PQW}$$

- $PQW \Leftrightarrow RSW$
  - $$\begin{aligned} \vec{r}_{RSW} &= R_3(\nu) \vec{r}_{PQW} \\ \vec{r}_{PQW} &= R_3(-\nu) \vec{r}_{RSW} \end{aligned}$$

- $IJK \Leftrightarrow EQW$

$$\begin{cases} \vec{r}_{IJK} = R_3(-\Omega) R_1(-\iota) R_3(fr\Omega) \vec{r}_{EQW} \\ \vec{r}_{EQW} = R_3(-fr\Omega) R_1(i) R_3(\Omega) \vec{r}_{IJK} \end{cases}, \quad fr = \begin{cases} +1 & 0 \le i \le 90^\circ \\ -1 & 90^\circ < i \le 180^\circ \end{cases}$$

- $IJK \Leftrightarrow RSW$ ($IJK \Leftrightarrow PQW \Leftrightarrow RSW$)
  - $\vec{r}_{IJK} = R_3(-\Omega) R_1(-\iota) R_3(-u) \vec{r}_{RSW}, \quad u = \nu + \omega$
  - $\hat{R} = \frac{\vec{r}}{|\vec{r}|} \quad \hat{W} = \frac{\vec{r} \times \vec{v}}{|\vec{r} \times \vec{v}|}, \ \hat{S} = \hat{W} \times \hat{R}$

$$\vec{r}_{IJK} = \begin{bmatrix} \hat{R} & \hat{S} & \hat{W} \end{bmatrix} \vec{r}_{RSW}$$

- $IJK \Leftrightarrow NTW$ ($IJK \Leftrightarrow PQW \Leftrightarrow RSW \Leftrightarrow NTW$)
  - $\vec{r}_{IJK} = R_3(-\Omega) R_1(-\imath) R_3(-u) R_3(-\phi_{fpa}) \vec{r}_{NTW}$
  - $\hat{T} = \dfrac{\vec{v}}{|\vec{v}|}, \; \hat{W} = \dfrac{\vec{r} \times \vec{v}}{|\vec{r} \times \vec{v}|}, \; \hat{N} = \hat{T} \times \hat{W}$

    $\vec{r}_{IJK} = \begin{bmatrix} \hat{N} & \hat{T} & \hat{W} \end{bmatrix} \vec{r}_{NTW}$

3.4.3 Velocity & Accelerations

- $\begin{cases} \vec{r}_{IJK} = n_1 \hat{I} + n_2 \hat{J} + n_3 \hat{K} \\ \vec{v}_{IJK} = \dot{n}_1 \hat{I} + \dot{n}_2 \hat{J} + \dot{n}_3 \hat{K} \\ \vec{a}_{IJK} = \ddot{n}_1 \hat{I} + \ddot{n}_2 \hat{J} + \ddot{n}_3 \hat{K} \end{cases}$

- $\vec{r}_{ijk} = \zeta_1 \hat{i} + \zeta_2 \hat{j} + \zeta_3 \hat{k}$

$$\begin{aligned} \vec{v}_{ijk} &= \vec{v}_{origin} + \left(\frac{d\vec{r}_{ijk}}{dt} \right)^{ijk} + \vec{\omega}_{i/I} \times \vec{r}_{ijk} \\ &= \vec{v}_{origin} + \left(\dot{\zeta}_1 \hat{i} + \dot{\zeta}_2 \hat{j} + \dot{\zeta}_3 \hat{k} \right) + \left(\zeta_1 \dot{\hat{i}} + \zeta_2 \dot{\hat{j}} + \zeta_3 \dot{\hat{k}} \right) \end{aligned}$$

$$\begin{aligned} \vec{a}_{ijk} &= \vec{a}_{origin} + \left(\frac{d^2\vec{r}_{ijk}}{dt^2} \right)^{ijk} + \vec{\alpha}_{i/I} \times \vec{r}_{ijk} + \vec{\omega}_{i/I} \times \left(\vec{\omega}_{i/I} \times \vec{r}_{ijk} \right) + 2\vec{\omega}_{i/I} \times \left(\frac{dr_{ijk}}{dt} \right)^{ijk} \\ &= \vec{a}_{origin} + \left(\ddot{\zeta}_1 \hat{i} + \ddot{\zeta}_2 \hat{j} + \ddot{\zeta}_3 \hat{k} \right) + 2\left(\dot{\zeta}_1 \dot{\hat{i}} + \dot{\zeta}_2 \dot{\hat{j}} + \dot{\zeta}_3 \dot{\hat{k}} \right) + \left(\zeta_1 \ddot{\hat{i}} + \zeta_2 \ddot{\hat{j}} + \zeta_3 \ddot{\hat{k}} \right) \end{aligned}$$

  - tangential acceleration
  - centripetal acceleration
  - Coriolis' acceleration

$$\vec{r}_{IJK} = R\vec{r}_{ijk}$$

- $$\vec{v}_{IJK} = R\left(\frac{d\vec{r}_{ijk}}{dt}\right)^{ijk} + \dot{R}\vec{r}_{ijk}$$

$$\vec{a}_{IJK} = R\left(\frac{d^2\vec{r}_{ijk}}{dt^2}\right)^{ijk} + 2\dot{R}\left(\frac{d\vec{r}_{ijk}}{dt}\right)^{ijk} + \ddot{R}\vec{r}_{ijk}$$

3.4.4 $IJK \Leftrightarrow (\lambda, \phi)$

- Problem : given $\vec{r}_{IJK}$, what are $\phi_{gc}, \phi_{gd}, \lambda, h_{el}$?

  h_{el} : height of the satellite above the ref. ellipsoid

  - Core technique in determining ground tracks
  - Direct algebraic relationship between Cartesian & spherical coordinates is not applicable due to the Earth's oblateness
  - The difference between ϕ_{gc} & ϕ_{gd} can cause errors up to 20km

- Solution by Spherical Geometry

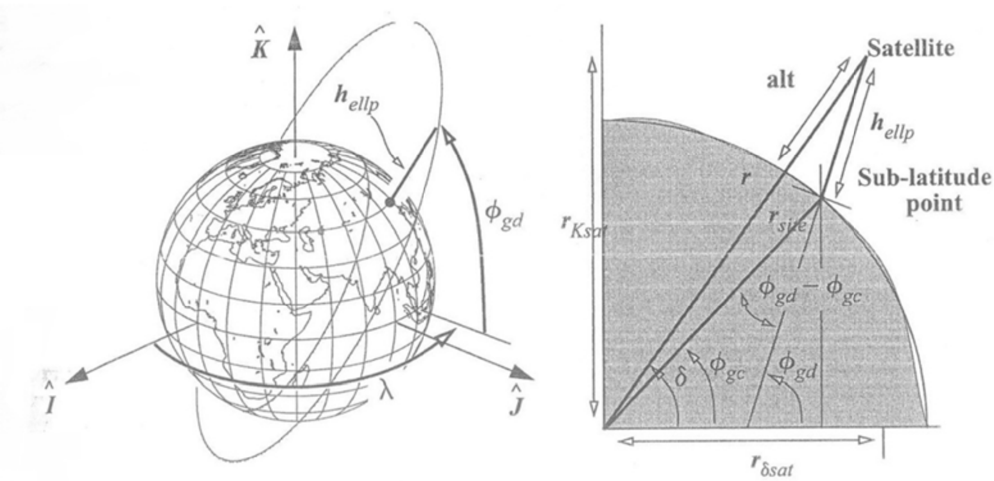

Figure 3-20. Determining a Satellite's Sub-latitude Point. Notice h_{ellp} is used for the altitude, and not H_{MSL}. The difference between the geodetic and geocentric angles is helpful in setting up the iteration for the problem.

- Obtain right ascension (α)

$$r_{\delta sat} = \sqrt{r_I^2 + r_J^2}$$
$$\sin\alpha = \frac{r_J}{r_{\delta sat}}$$
$$\cos\alpha = \frac{r_I}{r_{\delta sat}}$$

- Longitude $\lambda = \alpha - \theta_{GMST}$

- (Iterative) Equation for determining ϕ_{gd}

From (3.7)-(c, d)

$$\sin\phi_{gd} = \frac{r_K}{S_\oplus + h_{el}} \quad (A)$$
$$\cos\phi_{gd} = \frac{r_\delta}{C_\oplus + h_{el}} \quad (B)$$

From (A)

$$h_{el} = \frac{r_K}{\sin\phi_{gd}} - S_\oplus \quad (C)$$

From (3.7)-(a, b)

$$S_\oplus = \left(1 - e_\oplus^2\right) C_\oplus \quad (D)$$

$$\begin{aligned} \tan\phi_{gd} &= \frac{\sin\phi_{gd}}{\cos\phi_{gd}} \quad \Leftarrow (A),(B) \\ &= \frac{r_K}{r_\delta}\frac{C_\oplus + h_{el}}{S_\oplus + h_{el}} \quad \Leftarrow (C) \\ &= \quad \cdots \\ &= \frac{\left(C_\oplus - S_\oplus\right) S_{\phi_{gd}} + r_K}{r_\delta} \quad \Leftarrow (D) \end{aligned}$$

$$\tan\phi_{gd} = \frac{r_K + e_\oplus^2 C_\oplus \sin\phi_{gd}}{r_\delta}$$

Note that we do not know the site location (r_δ, r_K)

However, also note from Fig 3-21 that $(r_{\delta sat}, r_{Ksat})$ has similar geometric characteristics to (r_δ, r_K) in terms of ϕ_{gd}

So, we can use the above equation by replacing

$r_\delta \Leftarrow r_{\delta sat}$, $r_K \Leftarrow r_{Ksat}$.

Then we solve iteratively $tan\,\phi_{gd} = \dfrac{r_K + e_\oplus^2 C_\oplus \, sin\,\phi_{gd}}{r_\delta}$ for ϕ_{gd}

- Finally obtain h_{el} by rearranging (B)

$$h_{el} = \frac{r_{\delta sat}}{cos\,\phi_{gd}} - C_\oplus$$

- Refer to textbook for solution by analytic method

3.5. Time

- Newcomb: the main purpose of time is to define with precision the moment of phenomenon

- Epoch designates a particular instant of a phenomenon usually described as a ‘date’
- Determination of an epoch requires
  - A Fundamental Epoch
    - Example: beginning of the time era, Jan. 1, 4713 BC for Julian Date
  - Precise Repeatable Time Interval
    - Day
    - Hour/Minute/Second
    - Degree(°)/Arcminute(‘), Arcsecond(“)
    - 24 hours = 360° of longitude = equivalent time interval

- Time Interval Conversion

$$1^h = 60 \text{ minutes}(60^{\mathrm{m}}) = 3600 \text{ seconds}(3600^{\mathrm{s}})$$
$$1^\circ = 60 \text{ arcminutes}(60') = 3600 \text{ seconds}(3600'')$$

$$1^{\mathrm{h}} = 15^\circ \qquad 1^\circ = \frac{1^{\mathrm{h}}}{15} = 4^{\mathrm{m}}$$
$$1^{\mathrm{m}} = 15' \qquad 1' = \frac{1^m}{15} = 4^{\mathrm{s}}$$
$$1^{\mathrm{s}} = 15'' \qquad 1'' = \frac{1^{\mathrm{s}}}{15}$$

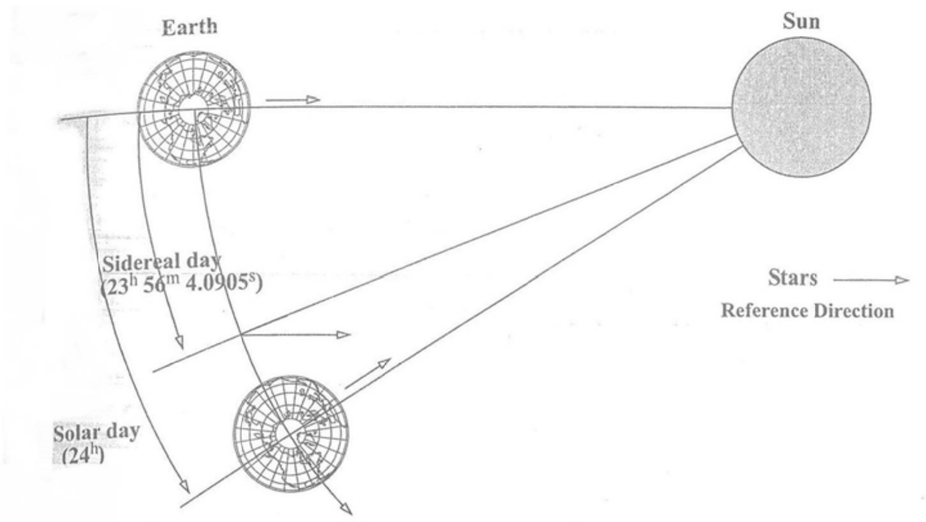

Figure 3-21. Measuring Solar and Sidereal Time. Distance and angular motion are greatly exaggerated for clarity. One solar day is the time required for an observer on the Earth to revolve once *and* observe the Sun at the same location. A sidereal day uses observations of the stars instead of the Sun. Notice that the sidereal day is slightly shorter than the solar day because the stars remain essentially in the same location over time.

- Four Main Time Scales
  - Solar Time (and Universal Time (UT))
    - Successive transits of the Sun over a local meridian
    - Universal Time (UT): UT0, UT1, UT2 based on a fictitious mean Sun exhibiting uniform motion
  - Sidereal Time
    - Successive transits of the Stars over a local meridian
    - 1 Solar Day = (1 Sidereal Day + about 1°)
      - due to the Earth revolution about the Sun
    - Solar Times and Sidereal Times are related through mathematical relations

- Dynamical Time
  - Measured by the motion of bodies
    - Example: Moon's motion about the Earth
  - Depends on the fact that time is an independent variable in the equations of motion describing the object's motion
  - Precise determinations requires relativistic corrections in modeling the Earth's motion
- Atomic Time
  - The most precise time standard
  - Based on specific quantum transition of electrons in a cesium-133 atom

3.5.1. Solar Time and Universal Time

- Solar Time is based on the interval between successive transits of the Sun over a local meridian

- Apparent Solar Time
  - Interval between successive transits which we observe from a particular longitude
    - Local Apparent Solar Time = $LHA_{\odot}$+12h
    - Greenwich Apparent Solar Time = θ_{GMST}-$\alpha_{\odot}$+12h
  - 12h increment ensures that 0h corresponds to midnight and 12h to noon, respectively
  - Not uniform due to:

◆ the eccentricity of the Earth's orbit

◆ 23.5° inclination of the ecliptic to the celestial equator;

the projection of the solar motion along the ecliptic onto the celestial equator appears sinusoidal

◆ The variation in the Sun's apparent motion in right ascension

■ Replaced by Mean Solar Time as the primary reference for time keeping

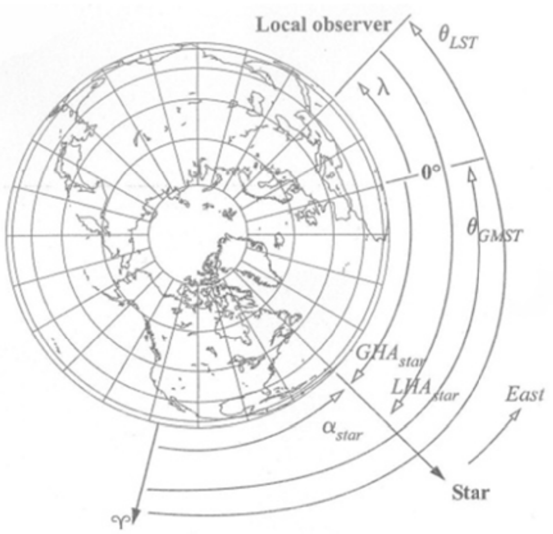

Figure 3-9. Geometry for LHA and GHA. GHA, and LHA are always measured positive to the west *from* the observer to the object of interest. Longitude, λ, is positive to the east. Both GHA_{star} and LHA_{star} are positive here. We use subscripts to avoid confusion when referring to specific objects. θ_{LST} and θ_{GMST} wiii be discussed shortly.

● Mean Solar Time & Universal Time (UT)

■ A Fictitious Mean Sun (FMS) proposed by Simon Newcomb

◆ Nearly uniform motion along the celestial equator

◆ $\alpha_{FMS} = 18^{h}\ 38^{m} 45.836^{s} + 8,640,184.542^{s} T + 0.0929^{s} T^{2}$

● T : Julian Centuries from Greenwich at noon Jan. 0, 1900

  - Neglecting Variations
    - In the local meridian due to polar motion
    - In the Earth's non-uniform rotation rate
- Universal Time
  - Defined as Mean Solar Time at Greenwich
  - Greenwich Mean Time (GMT), the local time in English is often incorrectly assumed to be UT
    - GMT ≠ UT
  - UT0
    - Difficult to measure the Sun's motion with enough precision
    - Indirectly determined by first measuring stars and then obtaining through math formula between Solar Time and Sidereal Time
    - For an observer at a known longitude:

      $$UT0 = 12^h + GHA_e = 12^h + LHA_e - \lambda$$

      GHA and LHA are positive to the West
  - UT1
    - Correction of UT0 for polar motion
    - Independent of station location
    - $UT1 = UT0 - (x_p SIN(\lambda) + y_p COS(\lambda)) TAN(\phi_{gc})$

      λ: longitude

      (x_p, y_p): instantaneous positions of the pole (which are discussed later)

      ϕ_{gc}: geocentric latitude of the observing site

- $|UT0-UT1| \sim 0.030^s$

◆ UT2

- Correction of UT1 for seasonal variations
- Considered obsolete

■ Equation of Time (EQ_{time})

◆ Difference between apparent and mean solar time

◆ Essentially difference between Sun's true right ascension and the right ascension of the FMS

$$EQ_{time} = -1.914\ 666\ 471^{\circ}\,\mathrm{SIN}(M_{e}) - 0.019\ 994\ 643\ \mathrm{SIN}(2M_{e}) + 2.466\ \mathrm{SIN}(2\lambda_{ecliptic}) - 0.0053\ \mathrm{SIN}(4\lambda_{ecliptic})$$

$EQ_{time} \in [-14^m, 16^m]$ during the year

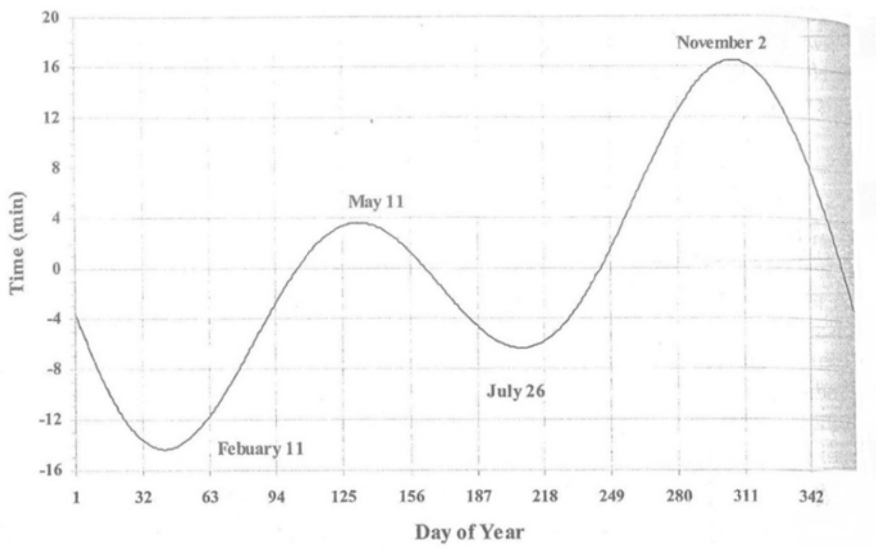

Figure 3-22. Equation of Time Variation. The difference in mean and apparent (or true) solar time varies throughout the year from about plus 16 minutes, to about minus 14 minutes.

- Conversion between Solar Time and Sidereal Time

  1 solar day = 1.002 737 909 350 795 sidereal day
  1 sidereal day = 0.997 269 566 329 084 solar day
  1 solar day = $24^{h}\ 3^{m}\ 56.555\ 367\ 8^{s}$ sidereal time
  1 sidereal day=$23^{h}\ 56^{m}\ 4.090\ 524^{s}$ solar time

  - Conversion between **Mean** Solar Time and **Mean** Sidereal Time

    - Differences occur because of non-uniform rotation of the Earth

      $$\omega_{\oplus_{prec}} = 1 \text{ solar day} = 1.002\ 737\ 909\ 350\ 795 + 5.9006\times10^{-11}T_{UT1} - 5.9\times10^{-15}T_{UT1}^{2} \text{ sidereal day}$$

      T_{UT1}: the number of Julian centuries since J2000

    - The value also approximates the mean rotation of the Earth w.r.t. a precessing equinox.

- Coordinated Universal Time (UTC)

  - Derived from Atomic Time
  - Designed to follow UT1 within ΔUT1 = |UT1-UTC| ≤ 0.9^{s}
  - Basis of civil time system on ordinary clock
  - Also called Zulu Time
  - Because UT1 varies irregularly due to variations in the Earth rotation, we must periodically insert "Leap Seconds' into UTC to keep two time scales in close agreement, i.e., UT1 ≤ 0.9^{s}
  - UTC always differs from TAI (Temps Atomique International) by an integer number of seconds

- Julian Date (JD)
  - Interval of time measured in days from the epoch Jan. 1, 4713 B.C. 12:00

$$ALGORITHM\ 14: JULIAN\ DATE\ (yr, mo, d, h, \min, s \Rightarrow JD)\ \{1900\ to\ 2100\}$$

$$JD = 367(yr) - INT\left\{\frac{7\left\{yr + INT\left(\frac{mo+9}{12}\right)\right\}}{4}\right\} + INT\left(\frac{275mo}{9}\right)$$

$$+ d + 1{,}721{,}013.5 + \frac{\frac{\left(\frac{s}{60*} + \min\right)}{60} + h}{24}$$

- Empirical Formula

  - Valid for any time system discussed; UT1, TDT, TDB, but usually implies a time system based on UT1 unless otherwise specified
  - 8 decimal digits ⇔ 4^{s}X10^{-4} accuracy
  - Modified Julian Date (MJD)

$$MJD = JD - 2{,}400{,}000.5$$

- Size of the date by about two significant days
- Begins each day at midnight (instead of noon)

  Note 0.5 in 2,400,000.5

  - Refer to textbook for Meeus' formula for JD without restriction
  - Some commonly used epochs:

J2000.0 = 2,451,545.0 = January 1, 2000 12:00 TT
J1900.0 = 2,415,021.0 = January 1, 1900 12:00 UT1
B1900.0 = 2,415,020.313 52 = December 31, 1899 19:31:28.128 UT1
B1950.0 = 2,433,282.423 459 05 = December 31,1949 22:09:46.862 UT1

- J: Julian

- B: Besselian: Old Systems

- JD contains (yr, mo, day, hour, min, sec) in one variable
  - Continuous, simple, concise
  - Good for computer applications
- Julian Centuries from a particular epoch (i.e., J2000)

$$T_{xxx} = \frac{JD_{xxx} - 2,451,545.0}{36,525}$$: xxx can be UT1, TT, ...

- Day of the Year
  - Days are recorded for the end of each month

TABLE 3-3. **Day of the Year.** This table lists numbers for days of the year. The elapsed days from the beginning of the year are often useful. Leap-year values are in parentheses.

Date (0^h)	Day of the Year	Elapsed Days from Jan 1, 0^h	Date (0^h)	Day of the Year	Elapsed Days from Jan 1, 0^h
January 1	1	0	July 1	182 (183)	181 (182)
January 31	31	30	July 31	212 (213)	211 (212)
February 28	59	58	August 31	243 (244)	242 (243)
March 31	90 (91)	89 (90)	September 30	273 (274)	272 (273)
April 30	120 (121)	119 (120)	October 31	304 (305)	303 (304)
May 31	151 (152)	150 (151)	November 30	334 (335)	333 (334)
June 30	181 (182)	180 (181)	December 31	365 (366)	364 (365)

3.5.2. Sidereal Time

- Sidereal Time
  - Defined as the hour angle of the vernal equinox relative to the local meridian
  - Time is an 'angle' measured from the observer's longitude to the equinox
  - Direct measure of the Earth's rotation

■ Greenwich Mean Sidereal Time (θ_{GMST})

Local Sidereal Time (θ_{LST})

$$\theta_{LST} = \theta_{GMST} + \lambda$$

◆ λ: (+)/(-) for east/west longitude

◆ $\theta_{GMST}, \theta_{LST}$: measured (+) in counterclockwise (CCW)

◆ GHA, LHA: measured (+) in clockwise (CW)

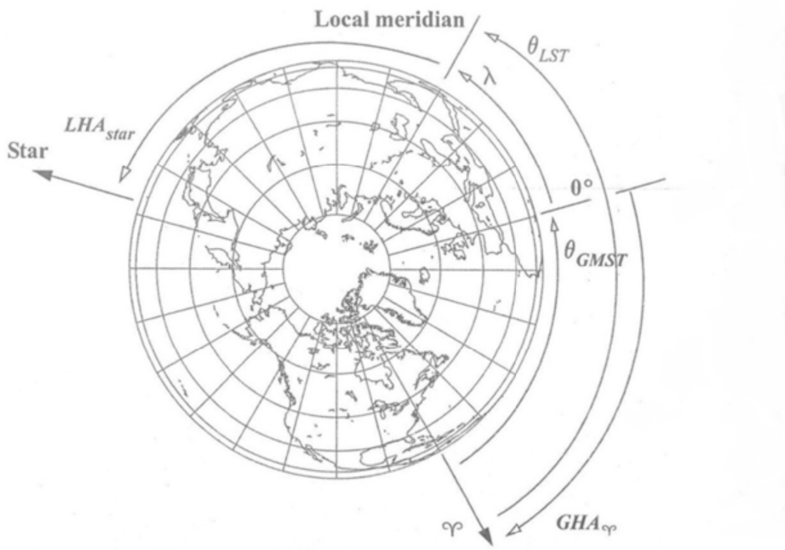

Figure 3-23. Geometry for LST and GMST. Sidereal time is measured positive to the east from the vernal equinox to the location of interest. If we try to locate a star, we can also use hour angles with an appropriate subscript (LHA_{star}). In this example, both sidereal times are positive in a right-handed system, whereas $GHA_{♈}$ is positive and LHA_{star} is negative in a left-handed system.

● Mean/Apparent Sidereal Time (MST/AST)

■ MST refers to a mean equinox that moves only with secular motion

AST refers to the true vernal equinox which includes secular and periodic motion

■ Equation of Equinoxes

- ◆ Difference between the mean equinox and the apparent equinox (in the plane of the true equator)

- ◆ Difference between MST and AST

- $$\theta_{LST} = \alpha_{star} + GHA_{star} + \lambda = \alpha_{star} + LHA_{star}$$
  $$LHA_{star} = GHA_{star} + \lambda$$

- Sidereal Time is equal to the Hour Angle of the fictitious mean Sun (FMS)

- $\theta_{GMST}, \theta_{LST} = f(JD_{UT1}, \lambda)$

  $$T_{UT1} = \frac{JD_{(UT1)} - 2{,}451{,}545.0}{36{,}525}$$
  $$\theta_{GMST} = 67{,}310.548\ 41^{s} + (876{,}600^{h} + 8{,}640{,}184.812\ 866^{s})T_{UT1} + 0.093\ 104\ T_{UT1}^{2} - 6.2 \times 10^{-6} T_{UT1}^{3}$$
  $$\theta_{LST} = \theta_{GMST} + \lambda$$

3.5.3. Atomic Time

- International Atomic Time =Temps Atomique International (TAI)
  - Highly accurate time system based on counting the cycles of a high-frequency electrical circuit maintained in resonance with a cesium-133 atomic transition
  - Precision to the extent that we observe relativistic effects for clocks in motion or accelerated by a local gravitational field

3.5.4. Dynamical Time

- Dynamical Time
  - Ephemeris Time (ET): obsolete
  - Terrestrial Time (TT): used to be Terrestrial Dynamical Time (TDT)
  - Barycentric Dynamical Time (TDB)
  - Geocentric Coordinate Time (TCG)
  - Barycentric Coordinate Time (TCB)
  - ➢ TCG and TCB for Space-Time applications within the framework of general relativity

- Terrestrial Time (TT)
  - Theoretical Time Scale of apparent geocentric ephemerides of bodies in the Solar System
  - Independent of Equations of Motion theories
  - SI second as the fundamental interval

$$
\begin{aligned}
UTC &= UT1 - \Delta UT1 \\
TAI &= UTC + \Delta AT \\
TT &= TAI + 32.184^{s}
\end{aligned}
$$

    - ◆ $\Delta AT, \Delta UT1$ are the Earth Orientation Parameters (EOP) that describe the motion of the Earth, and are accumulated differences
    - ◆ 32.184^{s} difference is the predefined constant offset between two systems

TABLE 3-4. Timing Coefficients. This table shows sample values for the difference in atomic time ($\Delta AT = TAI - UTC$) as well as coordinated and universal time ($\Delta UT1 = UT1 - UTC$). ΔAT remains constant until changed, whereas ΔUT1 changes continuously. Offsets to the GPS satellites are needed when determining navigational information. $GPS - UTC = \Delta AT - 19^s$. (Source: *Astronomical Almanac*, 2006:K9, and USNO)

Date		ΔAT (^{s}) (After Jan 1)	ΔUT1 $UT1 - UTC$ (^{s})	Date		ΔAT (^{s}) (After Jan 1)	ΔUT1 $UT1 - UTC$ (^{s})
1988	Jan 1	24	0.364 300	1997	Jan 1	30	–0.111 026 0
1989	Jan 1	24	–0.116 600	1997	Jul 1	31	0.526 894 0
1989	Jul 1	24	–0.386 170	1998	Jan 1	31	0.218 143 0
1990	Jan 1	25	0.328 640	1999	Jan 1	32	0.716 637 0
1991	Jan 1	26	0.618 670	2000	Jan 1	32	0.355 500 0
1992	Jan 1	26	–0.125 300	2001	Jan 1	32	0.093 227 6
1992	Jul 1	27	0.442 910 0	2002	Jan 1	32	–0.115 822 3
1993	Jan 1	27	0.062 020 0	2003	Jan 1	32	–0.289 318 0
1993	Jul 1	28	0.598 830 0	2004	Jan 1	32	–0.389 590 8
1994	Jan 1	28	0.199 280 0	2005	Jan 1	32	–0.503 902 4
1994	Jul 1	29	0.782 670 0	2005	Dec 31	32	–0.661 123 0
1995	Jan 1	29	0.398 500 0	2006	Jan 1	33	0.338 833 5
1996	Jan 1	30	0.555 290 0	2006	Jan 2	33	0.338 597 0

- Barycentric Dynamical Time (TDB)
  - Defined as the 'Independent Variable of the Equations of Motion with respect to the Barycenter of the Solar System'
  - TDB refers to the Barycenter of the Solar System

    TT refers to the Earth Center
  - $$TDB = TT + 0.001\ 658^s SIN(M_\oplus) + 0.000\ 013\ 85\ SIN(2M_\oplus)$$
    $$+\ \text{lunar/planetary terms} + \text{daily terms}$$

    $$M_\oplus \cong 357.527\ 723\ 3^\circ + 35{,}999.050\ 34\ T_{TDB}$$

    - $M_\oplus$: Mean Anomaly of the Earth's Orbit about the Sun
    - T_{TDB} : Julian Centuries of TDB
    - Note the iterative nature between TDB and T_{TDB}

- $TDB = TT + 0.001\ 658^{s} SIN(M_{\oplus}) + 0.000\ 013\ 85\ SIN(2M_{\oplus})$

  $M_{\oplus} \cong 357.527\ 723\ 3^{\circ} + 35{,}999.050\ 34\ T_{TT}$

  - T_{TT} : Julian Centuries of TT
  - Note the non-iterative nature between
  - Accuracy comparable to the above equation neglecting lunar/planetary terms

3.6. Time Conversions

- DMS ⇔ Rad

ALGORITHM 17: DMSTORAD(°, ', " ⇒ α)

$$\alpha = \left(\deg + \frac{'}{60} + \frac{''}{3600} \right) \frac{\pi}{180^\circ}$$

ALGORITM 18: RADTODMS (α ⇒ °, ', ")

$$Temp = \alpha \left(\frac{180^\circ}{\pi} \right)$$

$$\deg = \mathrm{TRUNC}(Temp)$$
$$' = \mathrm{TRUNC}((Temp - \deg)60)$$
$$'' = \left(Temp - \deg - \frac{'}{60} \right) 3600$$

- HMS ⇔ Rad

ALGORITHM 19: HMSTORAD(h,min,s ⇒ τ)

$$\tau = 15 \left(h + \frac{\min}{60} + \frac{s}{3600} \right) \frac{\pi}{180^\circ}$$

ALGORITHM 20: RADTOHMS(τ ⇒ h,min,s)

$$Temp = \tau \frac{180^\circ}{15\pi} = \tau \frac{24^{\mathrm{h}}}{2\pi}$$

$$h = TRUNC(Temp)$$
$$\min = TRUNC((Temp - h)60)$$
$$s = \left(Temp - h - \frac{\min}{60} \right) 3600$$

- HMS ⇔ Time of Day

ALGORITHM 21: TIMETOHMS($\tau \Rightarrow$ h,min,s)

$$Temp = \frac{\tau}{3600}$$

$$h = \mathrm{TRUNC}(Temp)$$
$$m = \mathrm{TRUNC}((Temp - h)60)$$
$$s = \left(Temp - h - \frac{\min}{60}\right)3600$$

- YMD ⇔ Day of Year
  - The only issue is to determine if a specific year is a leap one
    - Years evenly divisible by 4
    - Centuries evenly divisible by 400

$$\mathrm{IF}(Yr \text{ MOD } 4) = 0 \text{ THEN}$$
$$LMonth[2] = 29$$

- YMDHMS ⇔ Days
  - YMDHMS ⇒ Days

$$YMD\,TO\,DAY\,OF\,YEAR\,(YMD \Rightarrow Day\ of\ \ Yr)$$
$$days = day\ of\ \ Yr\ +\ h/24\ +\ \min/1440\ +\ s/86400$$

- Days ⇒ YMDHMS

$$Day\ of\ \ Yr = \mathrm{TRUNC}(Days)$$
$$YMDTODAYOFYEAR(YMD \Leftarrow Day\ of\ \ Yr)$$

$$\tau = (Days - Day\ of\ \ Yr)86{,}400$$
$$TIMETOHMS(\tau \Rightarrow h, \min, s)$$

- Julian Date to Gregorian Date
  - The process for finding the Julian date is straightforward in Algorithm 14.
  - But reversing it to find the Gregorian calendar date is also necessary.....

$ALGORITHM\ \ 22: JDTOGREGORIANDATE(JD \Rightarrow Year, Mon, Day, h, \min, s)$

$$T_{1900} = \frac{JD - 2{,}415{,}019.5}{365.25}$$

$Year = 1900 + \mathrm{TRUNC}(T_{1900})$

$LeapYrs = \mathrm{TRUNC}((Year - 1900 - 1)(0.25))$

$Days = (JD - 2{,}415{,}019.5) - ((Year - 1900)(365.0) + LeapYrs)$

IF $Days < 1.0$ THEN

$Year = Year - 1$

$LeapYrs = \mathrm{TRUNC}((Year - 1900 - 1)(0.25))$

$Days = (JD - 2{,}415{,}019.5) - ((Year - 1900)(365.0) + LeapYrs)$

IF $(Year MOD 4) = 0$ THEN

$LMonth[2] = 29$

$DayofYr = \mathrm{TRUNC}(Days)$

Sum days in each month until LMonth + 1 summation > *DayofYr*

$Mon = \#$ of months in summation

Day = *DayofYr* – *LMonth* summation

$\tau = (Days - DayofYr)24$

$h = \mathrm{TRUNC}(\tau)$

$\min = \mathrm{TRUNC}((\tau - h)60)$

$$s = \left(\tau - h - \frac{\min}{60}\right)3600$$

3.7. Transformation between Celestial and Terrestrial Coordinates

- Celestial=Inertial ⇔ Terrestrial = Body-Fixed Rotating
  - Related to each other by a series of translations and rotations known as Earth Orientation Parameters (EOP) = Reduction Formulas
    - Classical Transformation by a series of translation and rotation
    - Non-Rotating Origin, Celestial Intermediate Origin

- Ideal vs. True Axis of Rotation
  - ITRF is defined by convention, not by observation
  - International Reference Pole (IRP) is the location of terrestrial pole agreed upon by International Committees
  - Celestial Ephemeris Pole (CEP) is the (true) axis of rotation, which is normal to the true equator
  - The axes of ITRF/IRP and of CEP are slightly different (due to polar motion)
  - (ϕ, λ) is with respect to the CEP
  - The motion of CEP is roughly a circular spiral about IRP

- Polar Motion
  - Movement of the rotation axis with respect to the Earth crust
  - Change of bodies between IRP and CEP

- Represented by (x_p, y_p), a part of EOP
  - x_p : measure (+) south along 0° longitude meridian
  - y_p : meridian

TABLE 3-5. Pole Positions. This table shows the coordinates of the North Pole over about 5 years. Get current values for highly precise studies. All units are arcseconds ("). Remember that 0.1" = 0.000 028°, which is about 3.09 m for the surface of the Earth. x_p is measured along the 0° longitude meridian while y_p refers to the 90° W longitude meridian.

	Jan 1		Apr 1		Jul 1		Oct 1	
Year	x_p	y_p	x_p	y_p	x_p	y_p	x_p	y_p
2002	−0.1770	0.2940	−0.0307	0.5406	0.2282	0.4616	0.1985	0.2005
2003	−0.0885	0.1880	−0.1333	0.4356	0.1306	0.5393	0.2585	0.3040
2004	0.0313	0.1538	−0.1401	0.3206	−0.0075	0.5101	0.1985	0.4315
2005	0.1492	0.2380	−0.0287	0.2425	−0.0398	0.3968	0.0588	0.4169
2006	0.0527	0.3834	0.1028	0.3741	0.1285	0.2997	0.0327	0.2524

- The equatorial planes of the Earth and the Ecliptic are not actually fixed relative to the stars due to precession and nutation

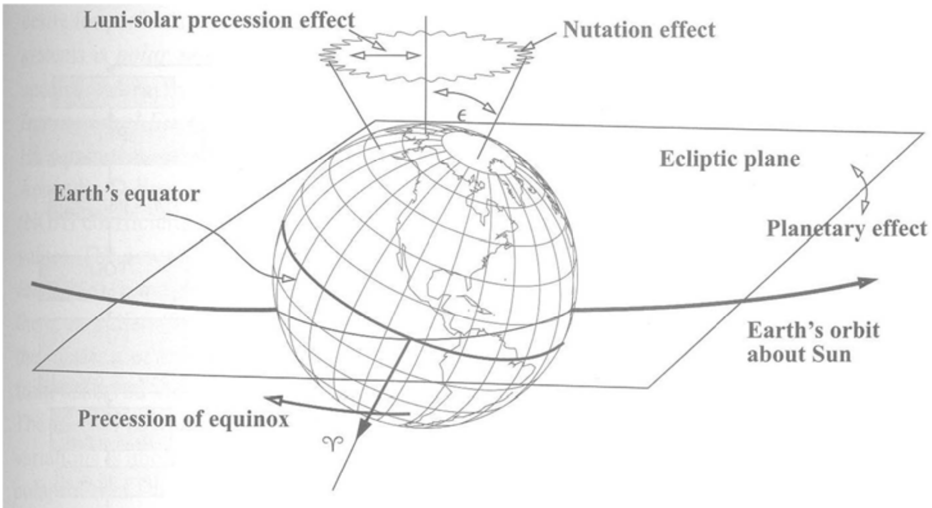

Figure 3-25. Precession and Nutation of the Earth's Equatorial Plane. This figure shows the combined effects of perturbing forces on the Earth. The general motion of the equinox moves west. Planetary and nutation effects are small compared to the luni-solar effects.

- Precession
  - Planetary Precession
    - Gravitational forces of the planets affect the Earth's orbit
    - Very slow secular change in the ecliptic's orientation
      - Equinox to precess to the West about 0.0033° (12")/century
      - Decreases the obliquity of ecliptic about 0.01305° (47")/century
  - Luni-Solar Precession
    - Sun and Moon's gravity produce a very small torque on the Earth because it is not completely spherical, i.e., mainly due to equatorial bulge
    - Results in a smooth wobbling
    - Period: 26,000 years; 0.013 846° (49.846")/year
  - Together, we call general precession; 0.013 889° (50")/year in longitude
  - Ecliptic of Date + Mean Equator of Date = Mean Equinox of Date (MOD)

- Nutation: Small Oscillations in the Earth's rotation axis
  - An additional torque the Moon produces on the Earth's equatorial bulge
    - Moon's orbital inclination of 5° about the Earth causes monthly variation in the periodic torque
    - Precession of the Moon's orbital plane with a period of 18.6 years due to solar perturbation

      = regression (the motion is in negative direction) of the lunar node
    - Advance of perigee of the lunar orbit due to solar perturbations

- Superimposition of all three effects
  - Many smaller periods
  - Maximum amplitude of about 0.0025° (9″) in the obliquity of the ecliptic
  - 0.00472° (17″) in longitude
- Mean Equinox of Date (MOD) + Nutation Effect

  ⇒ True Equator of Date (TOD)

3.7.1. Non-rotating Origin

(Not covered in class)

3.7.2. FK5 Reduction

(Not covered in class)

3.7.3. Precession (FK5 = J2000 Inertial)

- Transformation between FK5 and MOD

$$\begin{cases} \vec{r}_{MOD} = R_3(-z)R_2(\theta)R_3(-\zeta)\vec{r}_{FK5} \\ \vec{v}_{MOD} = R_3(-z)R_2(\theta)R_3(-\zeta)\vec{v}_{FK5} \end{cases}$$

$$z, \theta, \zeta = f(t) = \textbf{secular}$$

$$P(t) = R_3(-z)R_2(\theta)R_3(-\zeta)$$

3.7.4. Nutation

- Transformation between TOD and MOD

$$\begin{cases} \vec{r}_{TOD} = R_1(-\varepsilon)R_3(-\Delta\Psi)R_1(\bar{\varepsilon})\vec{r}_{MOD} \\ \vec{v}_{TOD} = R_1(-\varepsilon)R_3(-\Delta\Psi)R_1(\bar{\varepsilon})\vec{v}_{MOD} \end{cases}$$

$$\varepsilon, \Delta\Psi, \bar{\varepsilon} = f(t) = \textbf{periodic}$$

$$N(t) = R_1(-\varepsilon)R_3(-\Delta\Psi)R_1(\bar{\varepsilon})$$

3.7.5. Sidereal Time

- Transformation between TOD and PEF (Pseudo Earth Fixed)
  - TOD = non-rotating
  - PEF = rotating

$$\begin{cases} \vec{r}_{PEF} = R_3(\theta_{GAST})\vec{r}_{TOD} \\ \vec{v}_{PEF} = R_3(\theta_{GAST})\vec{v}_{TOD} \end{cases}$$

$$\theta_{GAST} = \theta_{GMST} + EQ_{equinox}$$

$$EQ_{equinox} = \Delta\Psi \cos\varepsilon + 0.00264'' \sin\Omega_{Moon} + 0.000063 \sin 2\Omega_{Moon}$$

$$ST(t) = R_3(\theta_{GAST}(t))$$

- Equation of Equinoxes ($EQ_{equinox}$) is the difference between θ_{GAST} and θ_{GMST}
  - Aligns the I-axis with the current site/satellite position
  - $EQ_{equinox} = \theta_{GAST} - \theta_{GMST}$ is the accumulated precession and nutation in right ascension

3.7.6. Polar Motion

- Transform between ITRF and PEF

$$\begin{cases} \vec{r}_{ITRF} = R_2\left(-x_p\right)R_1\left(-y_p\right)\vec{r}_{PEF} \\ \vec{v}_{ITRF} = R_2\left(-x_p\right)R_1\left(-y_p\right)\vec{v}_{PEF} \end{cases}$$

$$PM^T(t) @ R_2\left(-x_p\right)R_1\left(-y_p\right)$$

- Cannot predict values of $\left(x_p, y_p\right)$ with high accuracy over long periods

3.7.7. Summary (FK5)

- $$\begin{cases} \vec{r}_{J2000} = P(t)N(t)ST(t)PM(t)\vec{r}_{ITRF} \\ \vec{v}_{J2000} = P(t)N(t)ST(t)\left[PM(t)\vec{v}_{ITRF} + \vec{\omega}_{\oplus} \times \vec{r}_{ITRF}\right] \end{cases}$$

  - Need to reflect the Earth's rotation rate with respect to the Newtonian Inertial frame

  - $\vec{\omega}_{\oplus} = 7.29211514670698 \times 10^{-5}\left(1 - \dfrac{LOD}{86400}\right)$

    LOD = length of day

    = instantaneous rate of change (in seconds) of UTI with respect to a uniform time scale (UTC or TAI)

- $$\begin{cases} \vec{r}_{ITRF} = PM^T(t)ST^T(t)N^T(t)P^T(t)\vec{r}_{J2000} \\ \vec{v}_{ITRF} = PM^T(t)\dot{S}T^T(t)N^T(t)P^T(t)\vec{r}_{J2000} \\ \qquad + PM^T(t)ST^T(t)N^T(t)P^T(t)\vec{v}_{J2000} \end{cases}$$

  - $ST(t)$ is the only parameter that varies much

    $$\dot{S}T(t) = \begin{bmatrix} -\omega_{\oplus} sin\left(\theta_{AST}\right) & \omega_{\oplus} cos\left(\theta_{AST}\right) & 0 \\ -\omega_{\oplus} cos\left(\theta_{AST}\right) & -\omega_{\oplus} sin\left(\theta_{AST}\right) & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

- Both are equivalent to each other, though procedures are different from each other

- Implementation

$$
\begin{array}{ccc}
 & \boxed{ITRF} & \textit{True Earth Fixed} \\
PM(t) & \downarrow\uparrow & PM^T(t) \\
 & \boxed{PEF} & \textit{Pseudo Earth Fixed} \\
ST(t) & \downarrow\uparrow & ST^T(t) \\
 & \boxed{TOD} & \textit{True of Date} \\
N(t) & \downarrow\uparrow & N^T(t) \\
 & \boxed{MOD} & \textit{Mean of Date} \\
P(t) & \downarrow\uparrow & P^T(t) \\
 & \boxed{J2000} &
\end{array}
$$

- Computational Considerations
  - It is convenient to integrate the equations of motion in a Newtonian inertial frame; don't have to deal with Coriolis terms, etc.
  - J2000; mean equinox; mean equator is one of the representative inertial frames
  - Due to lengthy procedures of the complete FK5 reductions, there exist numerous approximations

3.7.8. FK4 reduction (B1950)

- B = Besselian year = time for the mean Sun's right ascension to increase by 24 hours
- $\vec{x}_{J2000}$; $[Known]\vec{x}_{B1950}$

3.7.9. Earth Models & Constants

- Canouical Units

$$DU = ER = R_{\oplus}$$

$$\mu_{\oplus} = \frac{DU^3}{TU^2} \quad \Rightarrow TU = \sqrt{\frac{DU^3}{\mu_{\oplus}}}$$

$$M = initial\ mass\ of\ spacecraft$$